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BREATHING APPARATUS WITH MASK ASSEMBLY FOR DELIVERING BREATHING GASES

Abstract

A mask frame includes a central conduit connection aperture and lateral arms having a 3-D curvature. The lateral arms extend outwardly from a center of the frame, rearwardly towards the patients ears, and upwardly along a vector passing from below the nose to a point between the temple the top of the ear. The lateral arms “twist” along their length, such that a bottom margin of an end of each lateral arm is positioned further away from a notional vertical plane passing through the centre of the conduit connection aperture than a top margin. An anti-rotation feature limits or prevents rotation between straps of a headgear and the mask frame.

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Background/Summary

INCORPORATION BY REFERENCE TO ANY PRIORITY APPLICATIONS [0001] This application is a continuation of U.S. application Ser. No. 17/447,757, filed Sep. 15, 2021, which is a continuation of U.S. application Ser. No. 15/524,802, filed May 5, 2017, now U.S. Pat. No. 11,154,681, which is a U.S. National Phase of PCT/NZ2015/050186, filed Nov. 6, 2015, which claims priority to U.S. Provisional Application No. 62/077,071, filed Nov. 7, 2014, and U.S. Provisional Application No. 62/212,431, filed Aug. 31, 2015, the entire contents of which applications are hereby incorporated by reference herein and made a part of the present disclosure.

BACKGROUND
Technical Field
[0002] The presently disclosed subject matter invention generally relates to interfaces for providing a supply of pressurised gas to a recipient.
Description of the Related Art
[0003] Breathing gases can be delivered to users with a variety of different mask styles and can be delivered for a variety of different purposes. For example, users can be ventilated using non-invasive ventilation (NIV). In addition, continuous positive airway pressure (CPAP) or variable airway pressure can be delivered using masks to treat a medical disorder, such as obstructive sleep apnea (OSA), chronic obstructive pulmonary disease (COPD), or congestive heart failure (CHF).
[0004] These non-invasive ventilation and pressure support therapies generally involve the placement of a user interface device, which is typically a nasal or nasal/oral mask, on the face of a user. The flow of breathing gas can be delivered from the pressure/flow generating device to the airway of the user through the mask.
[0005] Typically, patient interface devices include a mask frame that supports a sealing member. The sealing member contacts the facial surfaces of the user, including regions surrounding the nose, including the nose and the nares. Because such masks are typically worn for an extended period of time, a variety of concerns must be taken into consideration. For example, in providing CPAP to treat OSA, the user normally wears the mask all night long while he or she sleeps. One concern in such a situation is that the mask should be as comfortable as possible. It is also important that the mask provide a sufficient seal against a user's face without significant discomfort.
BRIEF SUMMARY
[0006] According to a first aspect of the invention there is provided a mask frame for a patient mask for delivering breathing gases to a patient, the mask frame comprising: [0007] a central region comprising a conduit connection aperture configured to be connected to a breathing gas delivery conduit, a notional central vertical plane extending through the centre of the conduit connection aperture; and [0008] first and second lateral arms each extending outwardly from the

central region away from the central vertical plane; [0009] each lateral arm having a length and terminating in a distal end remote from the central region, each lateral arm comprising a top and bottom margin; wherein [0010] each lateral arm twists along its length such that the bottom margin at the end of each lateral arm is positioned further away from the notional central vertical plane than the top margin at the end of each lateral arm.

[0011] The lateral arms may extend: [0012] laterally outwardly from the central region of the frame; [0013] rearwardly, towards the patients ears; and [0014] upwardly, so that the lateral arms are angled upwards such that they extend along a direction extending from the ends of the lateral arms to an area between the user's temples and ears. The lateral arms may extend upwardly along a vector passing from below the nose to a point between the temple at the top of the ear.

[0015] Each lateral arm may comprise a planar, generally oblong, strip or band, the end of each strip defining top and bottom corners at the top and bottom margins respectively, wherein the side arms twist along their length such that the bottom corner of the ends of the lateral arms are positioned further away from the central region of the frame than the top corners.

[0016] Each lateral arm may be tapered along its length, that is, the distance between the top and bottom margins reduces along at least part of the length of each lateral arm.

[0017] The ends of the lateral arms may be positioned below a notional horizontal mid plane that passes through the centre of the elbow connection aperture.

[0018] The end of each arm may comprise a headgear connector configured to connect the frame to headgear. The headgear connector may comprise a loop and a post configured to provide a connection point for a hook of a headgear clip attached to headgear.

[0019] In some embodiments, the headgear connector may comprise a rotation limiting formation configured to limit relative rotation between headgear connected to the headgear connector, and the mask frame. The rotation limiting formation may comprise an end stop against which the headgear abuts after a predetermined amount of relative rotation between the mask frame and headgear. The rotation limiting formation may be provided on the post. The rotation limiting formation may comprise two spaced apart end stops, relative movement between the headgear and the mask frame being limited by the distance between the two spaced apart end stops.

[0020] According to another aspect of the invention there is provided a mask assembly comprising the mask frame of the first aspect, and further comprising: [0021] headgear configured to be connected to the lateral arms of the mask frame; and [0022] a scaling cushion configured to be mounted on the mask frame.

[0023] According to a further aspect of the invention there is provided a mask assembly comprising a mask frame and a headgear configured to be connected to the mask frame, at least one of the mask frame and headgear comprising a connector comprising a post, the other of the mask frame and headgear comprising a connector comprising a hook configured to receive the post to connect the headgear to the mask frame such that the hook can rotate about the post towards and away from the mask frame, the mask assembly further comprising at least one rotation limiting formation configured to limit the extent of relative rotation between the hook and post.

[0024] The rotation limiting formation may comprise an end stop on one of the mask frame and headgear against which the other of the mask frame and headgear abuts after a predetermined amount of relative rotation between the mask frame and headgear. Two end stops may be provided, one on the mask frame, the other on the headgear, the end stops being configured to abut after a predetermined amount of relative rotation between the mask frame and headgear. One end stop may be provided on the hook, and the other end stop may be provided on the post.

[0025] At least one of the hook and post may comprise two spaced apart end stops, relative movement between the hook and post being limited by the distance between the two spaced apart end stops. One of the end stops may comprise a protrusion projecting from one of the hook and post, the other of the hook and post comprising a groove or recess having opposed ends, the opposed ends forming the two spaced part apart end stops, the protrusion being received in the

groove or recess when the headgear is connected to the mask frame and being configured such that the protrusion moves within the groove or recess, between the opposed ends. The groove or recess may be provided on the post, the protrusion being provided on the hook.
[0026] According to another aspect of the invention there is provided a headgear connector assembly for connecting a headgear to a mask, the headgear connector assembly comprising a first connector comprising a post, and a second connector comprising a hook configured to receive the post to connect the first connector to the second connector such that the hook can rotate about the post, the headgear connector assembly further comprising at least one rotation limiting formation configured to limit the extent of relative rotation between the hook and post.
[0027] According to a further aspect of the invention there is provided a mask assembly comprising a mask frame and a headgear configured to be connected to the mask frame, the mask frame comprising: [0028] a central region comprising a conduit connection aperture configured to be connected to a breathing gas delivery conduit, a notional central vertical plane extending through the centre of the conduit connection aperture; and [0029] first and second lateral arms each extending outwardly from the central region; [0030] each lateral arm having a length and terminating in a distal end remote from the central region, each arm comprising a top and bottom margin; wherein [0031] each lateral arm twists along its length such that the bottom margin at the end of each lateral arm is positioned further away from the notional central vertical plane than the top margin at the end of each lateral arm; [0032] the headgear comprising straps having headgear connectors configured to be rotatably connected to the ends of the lateral arms; [0033] the mask assembly further comprising at least one rotation limiting formation configured to limit the extent of relative rotation between the mask frame and the headgear.
[0034] According to another aspect of the invention there is provided a buckle for a closed loop headgear, the buckle comprising: [0035] an inside face and an outside face extending between a bottom face and a top face; [0036] an inside post, a central post, and an outside post, each post extending between the top and bottom faces; [0037] a friction loop opening positioned between the inside post and the central post and configured to receive part of a headgear strap loop; [0038] a front opening formed at least on the inside face; [0039] an outside opening formed at least on the outside face; [0040] a headgear strap path extending between the front opening and the outside opening between the outside post and the central post, the path being configured to receive a return part of the headgear strap loop, [0041] wherein the front opening and the outside opening are offset such that the headgear strap path is angled.
[0042] The buckle may further comprise a strap attachment surface located on an inside surface of the outside post along the headgear strap path, the strap attachment surface being configured to be attached to a strap of the headgear.
[0043] The front opening may be formed adjacent the bottom face of the buckle, such that the front opening extends through both the bottom face and the inside face.
[0044] The outside opening may be formed a distance along the outside face so as to be spaced from the bottom face.
[0045] The friction loop opening may comprise top and bottom margins at least one of which is configured to frictionally engage the headgear strap loop by virtue of the angle of the return part of the headgear strap through the angled path.
[0046] According to a further aspect of the invention there is provided a closed loop headgear, comprising: [0047] a buckle according to the aspect of the invention above; [0048] a rear headgear section; [0049] at least one strap extending forward from the rear headgear section, wherein the at least one strap passes through the friction loop opening between the inside post and the central post.
[0050] The strap preferably continues, passing through a hook connector of the headgear and doubles back to form the return part of the headgear, passing back through the angled headgear strap path.

[0051] The strap may include a grip strap end that extends through the buckle from the outside opening. The grip strap end may have a length long enough to grasp between a thumb and one or two fingers. A dimple may be provided for additional grip at a distal end of the grip strap end.
[0052] The rear headgear section may comprise a bifurcated rear section.
[0053] According to a further aspect of the invention there is provided an elbow for connecting a breathing gas delivery conduit to a patient mask assembly, the elbow comprising: [0054] a ball joint section and a bottom portion; [0055] an elongate tube section connecting the ball joint section and the bottom portion; wherein [0056] the elongate tube section is narrower than the bottom portion and a largest diameter of the ball joint section.
[0057] The elongate tube section may include a front section configured to face away from the user, in use, and a rear section configured to face towards the user, in use, the front section being longer than the rear section.
[0058] The elongate tube section may form a truncated top where the front section and rear section connect to the ball joint section.
[0059] The elongate tube section may form an elbow having an angle θ between a longitudinal axis of the elongate tube section, and a longitudinal axis of the ball joint section.
[0060] The front section of the elongate tube section may include a plurality of bias flow holes configured to enable CO2 washout during use. The plurality of bias flow holes may be arranged in columns along a length of the front section, and in one embodiment the plurality of bias flow holes is arranged in two arrays of two columns with a space between the two arrays. The columns within each section may be offset, such that the bias flow holes of each row are nested with respect to each other, that is, the bias flow holes of one column are at least partially located in spaces between the bias flow holes of an adjacent column. One column may have a different number of bias flow holes from another column.
[0061] The elongate tube section may include side sections, each having a notch. The notches may be configured to receive a feature of a diffuser body having corresponding geometry.
[0062] In other aspects of the invention there is provided: [0063] A: An apparatus as shown and described. [0064] B: A mask frame, comprising: side arms having a 3-D curvature. The side arms may extend: outwardly from a center of the frame, rearwardly, towards the patients ears, and upwardly, along a vector passing from below the nose to a point between the temple the top of the ear. The side arms may twist along their length, such that bottom corner of the ends of the side arms are positioned further away from a central portion of the frame than the upper corners. [0065] C: An elbow, comprising: [0066] a ball section and a wide bottom portion; [0067] an elongate tube section connecting the ball section and the wide bottom portion; [0068] the elongate tube section is narrower than the wide bottom end and a largest diameter of the ball section. The elongate tube section may include a long front section (facing away from the user and mask, in use) and a short rear section (facing towards the user and mask frame, in use). The elongate tube section may form a truncated top where the long front section and short rear section connect to the ball section. The elongate tube section generally forms an elbow having an angle θ. The long front section of the elongated tube section may include a plurality of bias holes for CO2 washout during use. The plurality of bias holes may be arranged in columns along a length of the long front section. The plurality of bias holes may be arranged in two sections of two columns with a space between the two sections. The columns within each section may be offset, such that the bias holes of each row are nested with respect to each other. The elongated tube section may include side sections, each having a notch. The notches may be configured to receive a corresponding geometry of a diffuser body. [0069] D: A buckle for a closed loop headgear, the buckle comprising: [0070] a bottom side and a top side; [0071] an inside post, a central post, and an outside post, each extending between the top and bottom sides; [0072] a friction loop opening between the inside post and the central post; [0073] a path comprising a front opening and an outside opening between the outside post and the central post, [0074] wherein the front opening and the outside are offset, forming an angled

path. A strap attachment surface may be located on an inside of the outside post. [0075] E: A closed loop headgear, may be provided comprising a buckle according to aspect D above and further comprising: [0076] at least one strap extending forward from a rear section passes through the buckle between the inside post and the central post. The strap may continue, passing through the hook connector and doubles back, passing back through the central post and outside post of the buckle. The strap may include a grip strap end that extends through the buckle from the outside opening. The grip strap end may have a length long enough to grasp between a thumb and one or two fingers. A dimple may be provided for additional grip at a distal end of the grip strap end. The rear section may be a bifurcated rear section.
[0077] Various features, aspects and advantages of the present invention can be implemented in any of a variety of manners. For example, while several embodiments will be described herein, sets or subsets of features from any of the embodiments can be used with sets or subsets of features from any of the other embodiments.
[0078] The term "comprising" is used in the specification and claims, means "consisting at least in part of". When interpreting a statement in this specification and claims that includes "comprising", features other than that or those prefaced by the term may also be present. Related terms such as "comprise" and "comprises" are to be interpreted in the same manner.
[0079] In this specification where reference has been made to patent specifications, other external documents, or other sources of information, this is generally for the purpose of providing a context for discussing the features of the invention. Unless specifically stated otherwise, reference to such external documents is not to be construed as an admission that such documents, or such sources of information, in any jurisdiction, are prior art, or form part of the common general knowledge in the art.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0080] These and other features, aspects and advantages of the present invention will be described with reference to the following drawings.
[0081] FIG. **1** is a top, front, and left side perspective view of a non-limiting exemplary embodiment of a nasal respiratory mask configured to provide a flow of pressurized breathing gases directly to a user's nares.
[0082] FIG. **1***a* is a front view of a flexible gas supply conduit, elbow assembly and swivel assembly that can be used with the mask of FIG. **1**.
[0083] FIG. **1***b* is a side view of the flexible gas supply conduit, elbow assembly and swivel assembly of FIG. **1***a*.
[0084] FIG. **1***c* is an enlarged perspective view of the elbow assembly and the flexible gas supply conduit of FIGS. **1***a* and **1***b*.
[0085] FIG. **2***a* is a top, front, and left side perspective view of a non-limiting exemplary embodiment of a sealing cushion that can be incorporated into the mask of FIG. **1**.
[0086] FIG. **2***b* is a bottom, rear, and right side perspective view of the sealing cushion of FIG. **2***a*.
[0087] FIG. **3***a* is a top, front, and left side perspective view of a non-limiting exemplary embodiment of a mask frame comprising a proximal side, a distal side, a central region, and a lateral arm that extends from each side of the central region and which can be incorporated with the sealing cushion of FIGS. **2***a* and **2***b* into the mask of FIG. **1**.
[0088] FIG. **3***b* is a bottom, rear, and right side perspective view of the mask frame of FIG. **3***a*.
[0089] FIG. **4** shows an orthogonal front view of the mask frame of FIGS. **3***a*-**3***b*.
[0090] FIG. **5** shows a corresponding orthogonal top view of the mask frame of FIGS. **3***a*-**3***b*.
[0091] FIG. **6** shows a corresponding orthogonal side view of the mask frame of FIGS. **3***a*-**3***b*.

[0092] FIG. **7** is a side view of a respiratory mask in use on a user U.
[0093] FIGS. **8** and **8**A-**8**F show a non-limiting exemplary embodiment of an elbow that can be incorporated into the mask of FIG. **1**, optionally with the sealing cushion of FIGS. **2***a*-**2***b* and the mask frame of FIGS. **3***a*-**3***b*.
[0094] FIGS. **9**A-D show a non-limiting exemplary embodiment of a diffuser configured to connect to the elbow of FIGS. **8** and **8**A-**8**F.
[0095] FIGS. **10**A-**10**E show the elbow of FIGS. **8** and **8**A-**8**F and diffuser assembly, wherein the diffuser is connected to the elbow such that the bias holes are covered.
[0096] FIGS. **11**A-**11**B show a non-limiting exemplary embodiment of a mask assembly with a closed loop headgear.
[0097] FIGS. **12**A-**12**F show a non-limiting exemplary embodiment of a buckle for use in a closed loop headgear.
[0098] FIGS. **13**A-**13**B are external perspective views of the buckle and strap used in the closed loop headgear.
[0099] FIGS. **14**A-**14**B the headgear strap passes through the buckle in cross-section.
[0100] FIGS. **15**A-**15**B show the closed loop headgear in sliding and friction modes.
[0101] FIGS. **16**A-**16**F show a non-limiting exemplary embodiment of a headgear connector assembly for connecting a headgear to a mask.

DETAILED DESCRIPTION

[0102] As used herein the term “retaining forces” refers to any force applied by a headgear to retain a respiratory mask on a user's face.
[0103] FIG. **1** shows a non-limiting exemplary embodiment of a nasal respiratory mask **1** configured to provide a flow of pressurized breathing gases directly to a user's nares. The respiratory mask **1** comprises a sealing cushion **3**, mask frame **5**, elbow assembly **7**, flexible gas supply conduit **9**, swivel assembly **11**, headgear **13** and a pair of headgear clips **15**.

Elbow and Flexible Gas Supply Conduit

[0104] With reference to FIGS. **1***a* to **1***c* an elbow assembly is configured to connect to an elbow connection aperture **77** of the mask frame **5** (see e.g., FIGS. **1** and **3***a*). The elbow assembly **7** includes a ball joint section **21** and a wide bottom portion **23**. An elongate tube section **25** connects the ball joint section **21** and the wide bottom portion **23**. In this example, the elongate tube section **25** is narrower than the wide bottom portion **23** and a largest diameter of the ball joint section **21**. The elongated tube section **25** includes a long front section **27** (facing away from the user and mask **1**, in use) and a short rear section **29** (facing towards the user and mask frame **5**, in use). The elongated tube section **25** forms a truncated top **31** where the long front section **27** and short rear section **29** connect to the ball joint section **21**. In this configuration, the elongated tube section **25** generally forms an elbow having a bend along its length, such that a longitudinal axis of a conduit receiving section to the elbow assembly **7** is not aligned with a longitudinal axis of the ball joint section **21**.
[0105] The long front section **27** of the elongated tube section **25** includes a plurality of bias flow holes **35** for CO2 washout during use. The plurality of bias flow holes **35** is arranged in columns along a length of the long front section, the columns generally being aligned with the longitudinal axis of the elongated tube section **25**. In particular, in this example, the plurality of bias flow holes **35** is arranged in three central longer columns and two lateral shorter columns with a space between the two sections. Each long central column has twelve bias flow holes in this example, and each short lateral column has seven bias flow holes. The columns are arranged on a tapering, keyhole shaped region of the long front section **27**.
[0106] The wide bottom portion **23** is a short tube section that is concentrically offset from a conduit receiving section, such that an annular channel is formed between the two. The conduit receiving section is configured to receive the flexible gas supply conduit **9** on its outer surfaces. The annular channel is configured to receive and retain the end of the flexible gas supply conduit **9**.

The conduit receiving section includes an external thread, which is configured to retain the flexible gas supply conduit **9**. The wide bottom portion **23** is configured to hide the end of the flexible gas supply conduit **9**.

[0107] The flexible gas supply conduit **9** is a flexible tube comprising an external helical bead **41** and a thin conduit wall **43**, supported by the bead **41**. The bead **41** may contain an electrically conducting wire or strip for providing heating to the conduit **9**, or for transferring data or sensor signals along the conduit **9**. The thin wall **43** comprises a radially outwardly directed fold or bend **45** between each adjacent helical coil. This outwardly directed fold **45** does not project outwardly as far as the bead **41**, such that the fold diameter is less than the bead diameter, when the conduit **9** is in a neutral condition, as shown in FIGS. **1***a* to **1***c*. The bead **41**, and fold **45**, are configured such that the conduit **9** is flexible such that it can bend and can extend or contract longitudinally. The fold **45** allows the conduit **9** to bend such that the conduit **9** is not straight in use. The fold **45** also allows the length of the conduit **9** to be varied. The bead **41** may be configured as a spring such that any flexing, extending or contracting of the conduit **9** is clastic, with the bead **41** and/or the fold, returning the conduit **9** to its neutral condition, in the absence of force applied the to the ends of the conduit **9**.

Scaling Cushion

[0108] The sealing cushion **3** is configured to engage with and form a substantially airtight seal with the nares and outwardly facing surfaces of a user's nose, such that pressurized breathable gases are delivered directly to the nasal passage. Such a seal is described in PCT/NZ2014/000150 (publication number WO2015009172), filed 17 Jul. 2014, the entire contents of which are hereby incorporated by reference.

[0109] The sealing cushion **3** preferably comprises a seal body **51** and a mask frame connector **53**. The seal body **51** can be formed of a soft and flexible material such that a supple pocket or envelope, that defines an inner cavity, is provided. The seal body **51** can be made of any appropriate material such as, but not limited to latex, vinyl, silicone or polyurethane.

[0110] Referring to FIGS. **2***a* and **2***b* the seal body **51** comprises a central portion **55** and first and second lateral portions **57** extending from each side of the central portion **55**. The seal body **51** further comprises an internal side **59** (FIG. **2***b*) and an external side **61** (FIG. **2***a*). The internal side **59** is positioned proximal to the user's face in use and is configured to provide sealing, stabilizing and locating surfaces. The external side **61** is positioned distal to the user's face, relative to the internal side **59**, in use, and is configured to provide structure to the scaling cushion **3**.

[0111] The internal side **59** of the central portion **55** is configured to extend across a base of a user's nose and the internal side of each of the lateral portions **57** is configured to curve around and extend across a lateral side of the nose. These lateral portions **57** can form a perimeter seal on outwardly facing surfaces or flanks of the nose. The lateral portions **57** are outwardly flared away from the user's nose at the lower corners, such that they contact the user's cheeks without digging in. The contact between the lateral portions **57** and the user's cheeks provides a location through which retaining forces can be applied by the headgear **13** to the user's face, in order to stabilize the respiratory mask **1**.

[0112] The internal side **59** has a thin wall thickness such that the seal body **51** is supple and capable of conforming to the geometry of the user's nose. The external side **61** comprises a greater wall thickness than the internal side **59**, such that it provides structure and stability to the more supple internal side **59**.

[0113] The internal side **59** of the central portion **55** further comprises a pair of prongs **62**. The prongs **62** comprise air delivery openings **63** and locating surfaces **65**. The air delivery openings **63** are configured to allow a flow of pressurized breathable gases to pass from within the seal body **51** to the user's airways. The locating surfaces **65** are configured to provide means of locating and sealing the prongs **62** within the user's nares and positioning the sealing cushion **3** on the nose.

[0114] The mask frame connector **53** is located within the external side **61** of the central portion **55**, and comprises a substantially rigid ring that is permanently attached to the seal body **51**, by any appropriate means. The mask frame connector **53** is configured to provide an inlet through which pressurized breathable air is delivered into the seal body **51**. The mask frame connector **53** is further configured to provide a substantially airtight connection between the sealing cushion **3** and the mask frame **5**; wherein the mask frame **5** and the sealing cushion **3** can be repeatedly assembled and disassembled. The connection between the mask frame **5** and the mask frame connector **53** may be achieved by any appropriate means including but not limited to snap-fit, friction fit, threaded or bayonet mechanisms.

Mask Frame

[0115] As shown in FIGS. **3***a* and **3***b* the mask frame **5** comprises a proximal side **71** (FIG. **3***b*), a distal side **73** (FIG. **3***a*), a central region **75**, and first and second lateral arms **97** that extend from each side of the central region **75**. The lateral arms **97** each comprise generally oblong planar strips or bands having elongate top and bottom margins extending along the length of each arm **97**. The lateral arms **97** are thus relatively wide when viewed from the front, but relatively narrow when viewed from above. The proximal side **71** is configured to be adjacent to the sealing cushion **3** and proximal to the user's face in use. The distal side **73** forms the external surface of the mask frame **5**. The distal side **73** of the central region **75** comprises an elbow connection aperture **77** which can optionally be circular. The elbow connection aperture **77** is configured to receive the ball joint **21** of the elbow assembly **17**. A planar surface **81** extends radially outwardly from the elbow connection aperture **77**.

[0116] The proximal side **71** of the central region **75** comprises an annular wall **83** that projects in a rearward direction towards the sealing cushion **3**, around the perimeter of the elbow connection aperture **77**.

[0117] The annular wall **83** comprises an internal surface **85** and an external surface **87**. The internal surface **85** comprises a concave spherical section, configured to form a ball joint socket. The ball joint socket is configured to connect to corresponding geometry on the elbow assembly **17**, namely the ball joint section **21**. The external surface **87** comprises one or more indentations **91** configured to be coupled to corresponding geometry on the mask frame connector **53** of the sealing cushion **3**, such that a connection between the mask frame **5** and the cushioning seal is achieved.

[0118] FIG. **4** shows an orthogonal front view of the mask frame **5**. The view is perpendicular to a front plane **80** that is coincident with the planar surface **81** (see e.g., FIG. **3***a*). The mask frame **5** is symmetrical about a central vertical plane ‘y’, which passed through a center of the elbow connection aperture **77**. The mask frame **5** further comprises a top edge margin **93** and a bottom edge margin **95**, both of which extend from one lateral arm **97** through the central region **75** to the other lateral arm **97**. The top edge margin **93** is positioned such that it is above the bottom edge margin **95** both in the view of FIG. **4** and in use. Within the central region **75**, the top edge margin **93** is offset from and substantially follows the curvature of the elbow connection aperture **77**. The top edge margin **93** tapers outwardly (horizontally away from the central vertical plane ‘y’) in a largely downward direction as it transitions from the central region **75** to each of the lateral arms **97**. The bottom edge margin **95** curves outwardly in a somewhat upward direction from the central vertical plane ‘y’ region. As it transitions from the ends of the central region **75** towards the end of each of the lateral arms **97** it curves in a slightly downward direction.

[0119] The tapered transitions of the top edge margin **93** and the bottom edge margin **95** form the lateral arms **97**, which are substantially elongate tapered members extending outwardly from each side of the central region **75**. The lateral arms **97** comprise a loop **99** and a post **101** configured to provide a connection point for the headgear clips (see e.g., FIG. **14**A). The loop **99** is a rectangular aperture that extends through the lateral arms **97** perpendicular to the distal side. In alternative embodiments the aperture may have any appropriate shape. The post **101** is formed by the outer wall of the loop **99** and forms the outer ends of the lateral arms **97**, and is substantially cylindrical. The post **101** is angled in an upwards direction towards the central region **77**. This is shown by the

width w.sub.1 of the bottom edge being wider than the width w.sub.2 of the top edge.
[0120] The ends of the lateral arms **97** are positioned substantially below a horizontal mid plane ‘x’ that is perpendicular to the front plane and vertical plane y and passes through the center of the elbow connection aperture **77**. Thus, the lateral arms **97** taper towards the bottom half of the mask frame **5**.
[0121] FIG. **5** shows a corresponding orthogonal top view of the mask frame **5**. It shows that the lateral arms **97** also extend in a rearward direction (distal to proximal) from the planar surface **81**.
[0122] FIG. **6** shows a corresponding orthogonal side view of the mask frame **5**. It can be seen that the end of the bottom edge margin **95** is closer to the front plane than the end of the top edge margin **93**. This results in the length of the post **101** forming an acute angle α with the front plane.
[0123] FIG. **7** is a side view of the respiratory mask **1** in use on a user U. It can be seen that, in use, the front plane is at an angle when the user U is upright. The angle of the posts **101** (noted above with reference to FIG. **6**) is such that the retaining force F applied by the headgear **13** is substantially perpendicular to the angle of the post **101**. This minimizes torsional forces between the headgear **13** and the mask frame **5**, and thus improves the stability of the respiratory mask **1** on the user U. The lateral arms **97** are angled upwards such they extend along a direction extending from the ends of the lateral arms **97** to an area between the user's temples and ears.
[0124] In other words, the lateral or side arms **97** have a 3-D curvature. In use, the lateral arms **97** extend: outwardly from the center of the frame **5**, rearwardly, towards the patients cars, and upwardly, along a vector passing from below the nose to a point between the temple the top of the car. Additionally, the lateral arms **97** “twist” along their length, such that bottom corners of the ends of the lateral arms **97** are positioned further away from the central vertical plane than the top corners, for example, as shown in FIG. **5**.
[0125] The positioning of the lateral arms **97** on the lower half of the mask frame **5** provides improved stability of the respiratory mask on the user's face. The retaining forces F applied by the headgear **13** to the mask frame **5** and sealing cushion **3** are predominantly applied through the lateral portions of the sealing cushion **3**. This results in the retaining forces being applied predominantly to the lower surfaces of the user's nose, their upper lip and cheeks. This provides a greater surface area over which the forces can be spread and thus greater stability. The spreading of the forces over a greater surface area may also minimize any pressure points on the user's face, which may cause discomfort. If the lateral arms **97** were located higher on the mask frame **5** the force vector F would be predominantly applied to the tip of the user's nose, which may provide reduced stability to the respiratory mask **1**. This may also cause discomfort to the user in the form of a pressure point on the tip of their nose.
[0126] The lower positioning of the lateral arms **97** provides the respiratory mask **1** with a more minimal appearance. Since the mask frame **5** is substantially positioned below the nose, in use, it is less visible and thus less dominating on the user's face. The lower positioning of the lateral arms **97** in combination with the angle of the post **99** also results in the headgear **13** passing over the user's cheek at a greater distance below the eyes. This is beneficial in that the headgear **13** is less likely to fall within the user's peripheral vision, making the mask **1** less obtrusive and more comfortable to wear.
Elbow and Diffuser
[0127] Shown in FIGS. **8** and **8**A-**8**F is an elbow **117** similar to the elbow **17** described above, and configured to connect to the elbow connection aperture **77** (see e.g., FIGS. **1** and **3***a*). The elbow **117** includes a ball joint section **121** and a wide bottom portion **123**. An elongate tube section **125** connects the ball joint section **121** and the wide bottom portion **123**. The elongate tube section **125** is narrower than the wide bottom portion **123** and a largest diameter of the ball joint section **121**. The elongated tube section **125** includes a long front section **127** (facing away from the user and mask, in use) and a short rear section **129** (facing towards the user and mask frame **5**, in use). The elongated tube section **125** forms a truncated top **131** where the long front section **127** and short

rear section **129** connect to the ball joint section **121**. In this configuration, the elongate tube section **125** generally forms an elbow having an angle Ø.

[0128] The long front section **127** of the elongate tube section **125** includes a plurality of bias flow holes **135** for CO2 washout during use. The plurality of bias flow holes **135** is arranged in columns along a length of the long front section **127**. In particular, as shown, the plurality of bias flow holes **135** is arranged in two sections or areas of two columns with a space between the two sections or areas of the elongate tube section **125**. The columns within each section or area are offset, such that the bias flow holes of each row are nested with respect to each other, that is, the bias flow holes of one column are nested at least partially within the spaces between the bias flow holes of another column. The elongated tube section **125** further includes side sections **137**, each having a notch **139**. The notches **139** are configured to receive a part of a diffuser body having corresponding geometry.

[0129] The wide bottom portion **123** is short tube section that is concentrically offset from a conduit receiving section **141**, such that an annular channel **143** is formed between the two. The conduit receiving section **141** is configured to receive the flexible gas supply conduit **9** on its outer surfaces. The annular channel **143** is configured to receive and retain the end of the flexible gas supply conduit **9**. The conduit **9** receiving section includes a retaining formation which may be a projection and which is in this example is a screw thread **145**, which is configured to retain the flexible gas supply conduit **9**. The wide bottom portion **123** is configured to hide the end of the flexible gas supply conduit **9**. Thus, the end of the flexible conduit **9** is threaded onto the elbow **117** to retain the conduit **9** on the elbow **117**.

[0130] Shown in FIGS. **9**A-D is a diffuser **151** configured to connect to the elbow **117**, covering the bias flow holes **135** such that gas exhausted from the bias flow holes **135** passes through the diffuser **151**. The diffuser **151** comprises a diffuser body **153** and a region or regions of a diffuser material **155**. The diffuser body **153** comprises a substantially rigid plastic component having two lateral sides **157** and a front region **159** there between. The diffuser body **153** is configured to have a curvature that compliments the curvature of the elongate tube section **125** of the elbow **117**, extending from one side section across the long front section to the other side section. The diffuser body **153** further comprises one or more diffuser openings **161** and a pair of grip tabs **163**. The diffuser openings **161** are located in the front region **159** such that they overlap the bias flow holes **135**, in use. In this example embodiment of the present disclosure the diffuser openings **161** are separated by a grill member **165**.

[0131] The grip tabs **163** extend approximately perpendicularly from the edges of each of the lateral sides **157** of the diffuser body **153**, and have a substantially trapezoidal profile. They are configured to allow the diffuser **151** to be gripped between the thumb and index finger of a user, during removal of the diffuser **151** from the elbow **117**.

[0132] The diffuser body **153** has an internal surface **167**, which is configured to be proximal to the elongate tube section **125** of the elbow **117**, in use. The internal surface **167** comprises a pair of engagement tabs **169**. The engagement tabs **169** extend inwardly from each of the grip tabs **163**, and are configured to engage with the notches **139** of the elbow **117**, such that the diffuser **151** is retained on the elbow **117**.

[0133] The regions of diffuser material **155** are configured to fill the diffuser openings **161**, providing a torturous path for exhaled air to pass through. The diffuser material **155** can be any breathable or porous material. In use, the diffuser material **155** is flush with the bias flow holes **135**, thus ensuring that air venting out of the bias flow holes **135** is diffused through the diffuser material **155**.

[0134] FIGS. **10**A-**10**D show the elbow **117** and diffuser **151**, wherein the diffuser **151** is connected to the elbow **117** such that the bias flow holes **135** are covered. It can be seen that, due to the narrow elongate tube section **125** of the elbow **117**, the diffuser **151** may now be configured to have a width that is substantially equal to the outer diameter of the wide bottom section **123** of the elbow

117. Additionally, the elongate tube section **125** of the elbow **117** allows for longer columns of bias flow holes **135**, and thus allows the elbow **117**, and therefore the elbow **117** and diffuser **151** to be narrower. This affords the elbow **117** and diffuser **151** assembly a smaller and less bulky appearance.

Headgear

[0135] As shown in FIGS. **1** and **7** the respiratory mask **1** is secured in place on a user's head by a headgear **13**. The headgear **13** comprises a three part construction, wherein the three parts include a pair of side straps **171** and a rear portion **173**. The side straps **171** are configured to join to the rear portion **173** at one end and be connected to a headgear clip **175** at the other end. The side straps **171** can be joined to the rear portion **173** by any suitable means including but not limited to ultrasonic welding, radiofrequency welding, adhesive(s) or sewing. The end of the side strap **171** that connects to the headgear clip **175** comprises a securement tab **177** for adjustment of the headgear size. The securement tab **177** comprises the hook element of a hook and loop connector such as but not limited to Velcro™. The securement tab **177** is configured to attach to the outer surface of the side strap **171** in order to secure the length of the side strap **171**, and thus the size of the headgear **13**. The rear portion **173** comprises a bifurcated component having an upper and lower strap **173***a*, **173***b*, configured to engage a user's head in the occipital region. The headgear **13** can be made from any suitable non-stretch or stretch material known in the art, including but not limited to Neoprene, TPE, Breathoprene™ or knitted fabric, or any combination of materials. In a preferred embodiment the rear portion **173** and side straps **171** are made from a single material in contrasting colours. In alternative embodiments the rear portion **173** can be made from a different material to the side straps **171**. In a further alternative embodiment the rear portion **173** and side straps **171** can be a single colour.

Closed Loop Headgear

[0136] FIGS. **11**A-**11**B show perspective and top views, respectively, of a non-limiting exemplary embodiment of a mask assembly **1** with a closed loop headgear **13**. The mask assembly **1** is attached to a bifurcated headgear **13**. Forward straps **171** extend forward from a rear, bifurcated section **173**. The forward straps **171** pass through a buckle **179** then pass through a hook **101** which is attached to the mask assembly **1**. Each forward strap **171** is doubled back, passing through the buckle **179**.

[0137] The headgear **13** shown in FIGS. **11**A-**11**B is a closed loop headgear, meaning that the length of the headgear can be adjusted from minimum length to a maximum length, but cannot be fully disassembled. A closed loop configuration may be desirable because the headgear **13** cannot be disassembled, and thus cannot be re-assembled incorrectly. The headgear **13** can be made from any suitable non-stretch or stretch material known in the art, including but not limited to Neoprene, TPE, Breathoprene™ or knitted fabric.

[0138] FIGS. **12**A-**12**F show a non-limiting exemplary embodiment of a buckle **179** for use in a closed loop headgear **13**, such as shown in FIGS. **11**A-**11**B. FIGS. **12**A-**12**B show and in an inside view and an outside view, respectively, of the buckle **179**. As shown, the buckle **179** includes a bottom face **181** and a top face **183**. Extending between the top and bottom sides **181**, **183** are an inside post **185**, a central post **187**, and an outside post **189**. Between the inside post **185** and the central post **187** is a friction loop opening **191**. Between the outside post **189** and the central post **187** is a path comprising a front opening **193** and an outside opening **195**. The front opening **193** and the outside opening **195** are offset, forming an angled path. Additionally, there is a strap attachment surface **197** located on an inside of the outside post **189**.

[0139] Shown in FIGS. **12**C-**12**D are perspective cross-sections of the buckle **179**, in particular to show internal features. On the central post **187** is a central post shoulder **199** and a central post recess **201**.

[0140] Shown in FIGS. **12**E-**12**F are front and back views, respectively of the buckle, in particular to show additional views of the above described features.

[0141] FIGS. **13**A-**13**B are external perspective views of the buckle **179** and strap **171** used in the closed loop headgear **13**. As shown, the strap **171** extending forward from the rear section **173** passes through the buckle **179** between the inside post **185** and the central post **187**. The strap **171** continues, passing through the hook (see e.g., **11**A) and doubles back, passing back through the central post **187** and outside post **189** of the buckle **179**. The strap **171** includes a grip strap end **205** that extends through the buckle **179**. The grip strap end **205** has a length long enough to grasp between a thumb and one or two fingers. A dimple **207** is provided for additional grip.
[0142] FIGS. **14**A-B show a cross-section of the strap **171** passing through the buckle **179**, forming an adjustable closed loop **209**. The strap **171** passes through the friction loop opening **191** formed between the inside post **185** and the central post **187**. The strap **171** passes back through the buckle **179**, through the angled path formed between the front opening **193** to the outside opening **195**.
[0143] As shown, the buckle **179** is relatively angled with respect to the strap **171**. As a result of the angle in which the strap **171** passes through the friction loop opening **191**, there is an interference corner **192** that generates a friction force that restricts the strap **171** from sliding though when under a tension. In particular, the plane of the strap **171** where the strap **171** passes through the friction loop opening is inclined relative to the plane of the buckle **179** so that the strap **171** is neither perpendicular nor parallel to the buckle **179**.
[0144] The strap **171** is permanently attached to the buckle **179** at a strap attachment surface **211** located on an inside of the outside post **189** (see e.g., FIG. **12**A). Because the strap **171** is permanently attached to the buckle **179**, the headgear **13** cannot be disassembled (i.e., the hooks on the adjustable closed loop cannot be removed). The strap **171** may be attached to the strap attachment surface **211** by any suitable means including but not limited to ultrasonic welding, radiofrequency welding, adhesive(s) or sewing.
[0145] FIGS. **15**A-B show the closed loop headgear **13** in sliding and friction modes, respectively. In FIG. **15**A a pull force **204** is applied to the grip strap end **205**. This pull force causes the buckle **179** to rotate **206** in the direction of the pull force (counter clockwise as shown). The "rotated" buckle **179** is represented in FIG. **15**A as a dashed rectangle. The buckle **179** rotates with the pull force because the strap **171** is attached to the buckle **179** at the strap attachment surface **211**. As the buckle **179** rotates, the buckle **179**, in particular the friction loop opening **191** becomes more perpendicular with the strap **171** sliding **208** through the opening **191**. As a result, the interference at the interference corner **192** (see e.g., FIG. **14**B), and therefore friction force, is decreased. This allows the strap **171** to pass freely (or at least with less resistance) through the friction loop opening **191**, thus allowing for adjustment.
[0146] Alternatively, as shown in FIG. **15**B, a tension force **210** is applied, for example, when the mask assembly **1** is in use. The tension force **210** causes the buckle **179** to rotate **202** in the direction of the tension force (clockwise as shown). As the buckle **179** rotates, the buckle, in particular, the friction loop opening **191** becomes more angled with respect to the strap **171** passing through the opening **191**. The "rotated" buckle **179** is represented in FIG. **15**B as a dashed rectangle. As a result, the interference at the interference corner (see e.g., FIG. **14**B) increases, and therefore the friction force between the friction loop opening **191** and the strap **171** passing through, increases.
Headgear Connector Assembly
[0147] With reference to FIGS. **16**A-**16**F, headgear **13** be connected to the mask frame **5** using a hook and post type headgear connector assembly **221**. In one example, the mask frame **5** comprises a post **101** on each lateral arm **97**. Each headgear clip **15** comprises a hook **15***a* configured to receive a respective post **101** of the mask frame **5**, to connect the headgear **13** to the mask frame **5**.
[0148] In one embodiment, each headgear clip **15** is able to rotate freely about the respective post **101** on the mask frame **5**. The post **101** is orientated generally vertically in normal use of the mask **1** and headgear **13**, and the side straps **171** therefore rotate laterally about a generally vertical pivot axis. This can have the effect that prior to the mask **1** and headgear **13** being placed on the head of

the user, the side straps **171** of the headgear **13** have pivoted around the posts **101** on the mask frame **5** to a position where part of the headgear **13** is adjacent the mask **1** and therefore impedes the mask I being located on the face of the user. The side straps **171** of the headgear **13** may therefore rotate around the posts **101** of the mask frame **5** such that the side straps **171** impede the internal side of the seal body **53**. This can make the mask **1** and headgear **13** difficult or inconvenient to put on.

[0149] It may be desirable to be able to prevent, limit or control the extent of relative rotational movement between the headgear **13** and the mask frame **5**, and particularly, to limit the extent of rotation of the side straps **171** of the headgear **13** relative to the mask frame **5**.

[0150] With further reference to FIGS. **16**A-**16**F, in one embodiment, the post **101** of the mask frame **5** and the hook **15***a* of the headgear clip **15** each comprise a respective movement limiting formation. These formations are arranged to engage to prevent, or at least limit the extent of, relative rotational movement between the mask frame **5** and the headgear **13**, and more particularly, between the mask frame **5** and the side straps **171** of the headgear **13**. The headgear connector assembly **221** between the mask **1** and headgear **13** therefore performs a rotation limiting function, to help prevent the side straps **171** of the headgear **13** rotating relative to the mask frame **5** to a position where they impede the mask **1**, and particularly the internal side of the seal body **53**.

[0151] In one embodiment each post **101** of the mask frame **5** comprises a recessed region, of reduced diameter as compared to the remainder of the post **101**, which forms a groove or recess **225** extending partially around the post **101**, in a plane perpendicular to the axis of the post **101**. When viewed from above, that is, along the axis of the post **101**, the groove or recess **225** extends around only a portion of the post **101**, that is, the groove or recess **225** is part circumferential. The groove or recess **225** may extend through about 180° for example. The groove or recess **225** thus comprises two opposed ends **225***a*, **225***b* where the groove or recess **225** meets the non-recessed part of the post **101**. These ends **225***a*, **225***b* are movement limiting formations comprising end stops.

[0152] Each hook **15***a* also comprises a movement limiting feature which in one example comprises a bump or protrusion **227** on the inside face of the hook **15***a* which functions as an end stop, that is, on the part of the hook **15***a* which receives the post **101** in use. When the post **101** is received in the hook **15***a*, which may be using a snap-fit type connection, relative rotation may occur between the post **101** and hook **15***a* with the bump **227** of the hook **15***a* moving within the groove or recess **225**. This allows the side straps **171** of the headgear **13** to pivot about the post **101** of the mask frame **5** to a limited extent. When a side strap **171** has rotated a predetermined distance relative to the mask frame **5**, the bump **227** reaches one end of the groove or recess **225** and abuts a groove or recess end stop **225***a*, **225***b*, this abutment preventing further relative rotation between the hook **15***a* and post **101**. The bump **227** can thus travel a predetermined distance along the groove or recess **225** before further movement of the bump **227**, and therefore rotation of the side strap **171** relative to the mask frame **5**, is prevented.

[0153] It will be appreciated that in another embodiment, the hook **15***a* and post **101**could be reversed, with the hook **15***a* being provided on the mask frame **5** and the post **101** being provided on the headgear clip **15**. Likewise, it is envisaged that the bump or protrusion **227** could be formed on the post **101**, and the groove or recess **225** on the hook **15***a*.

[0154] The groove or recess **225** may be provided at any location along the length of the post **101**, that is, at any axial position on the post **101**. More than one groove or recess **225** and bump **227** may be provided. The length of the groove or recess **225** and/or the size of the bump **227** may be selected to achieve the desired degree of relative rotation between the post **101** and hook **15***a*.

[0155] In at least one embodiment, the groove or recess **225** is positioned on a left side post **101** at a first vertical spacing and the groove or recess **227** is positioned on a right side at a second vertical spacing. In some embodiments the first and second vertical spacing are different. The corresponding left and right hooks **15***a* include a protrusion **225**, the protrusion **225** being at the

first vertical height and the second vertical height such that the left and right hooks **15***a* are connectable to one of the left side post **101** or the right side post **101**.
[0156] Unless the context clearly requires otherwise, throughout the description and the claims, the words “comprise”, “comprising”, and the like, are to be construed in an inclusive sense as opposed to an exclusive or exhaustive sense, that is to say, in the sense of “including, but not limited to”.
[0157] Reference to any prior art in this specification is not, and should not be taken as, an acknowledgement or any form of suggestion that that prior art forms part of the common general knowledge in the field of endeavour in any country in the world.
[0158] The inventions disclosed herein may also be said broadly to consist in the parts, elements and features referred to or indicated in the specification of the application, individually or collectively, in any or all combinations of two or more of said parts, elements or features.
[0159] Where, in the foregoing description reference has been made to integers or components having known equivalents thereof, those integers are herein incorporated as if individually set forth.
[0160] It should be noted that various changes and modifications to the presently preferred embodiments described herein will be apparent to those skilled in the art. Such changes and modifications may be made without departing from the spirit and scope of the inventions and without diminishing its attendant advantages. For instance, various components may be repositioned as desired. It is therefore intended that such changes and modifications be included within the scope of the inventions. Moreover, not all of the features, aspects and advantages are necessarily required to practice the present inventions. Accordingly, the scope of at least some of the present inventions is intended to be defined only by the claims that follow.

Claims

1. A mask frame for a patient mask for delivering breathing gases to a user, the mask frame comprising: a central region comprising a conduit connection aperture configured to be connected to a breathing gas delivery conduit, a notional central vertical plane extending through a center of the conduit connection aperture, a notional front plane extending radially outwardly from the conduit connection aperture; and a first lateral arm and a second lateral arm each extending outwardly from the central region away from the central vertical plane; wherein each of the first lateral arm and the second lateral arm have a length and terminate in a distal end remote from the central region, each of the first lateral arm and the second lateral arm comprising a top margin and a bottom margin; wherein the distal end of each lateral arm comprises a headgear connector comprising an aperture and a post configured to provide a connection point for headgear; and wherein an entirety of the bottom margin is closer to the front plane than an end of the top margin, such that the distal end of the post forms an acute angle with the front plane.
2. The mask frame of claim 1, wherein a planar surface extends radially outwardly from the conduit connection aperture, the front plane being coincident with the planar surface.
3. The mask frame of claim 2, wherein each of the first lateral arm and the second lateral arm extend in a rearward direction from the planar surface.
4. The mask frame of claim 1, further comprising a proximal side and a distal side, wherein the aperture extends through a respective one of the first lateral arm and the second lateral arm in a direction perpendicular to the distal side.
5. The mask frame of claim 4, wherein the post is formed by an outer wall of the aperture and forms the distal end of each of the first lateral arm and the second lateral arm.
6. The mask frame of claim 1, wherein the post is substantially cylindrical.
7. The mask frame of claim 1, wherein the post is at an angle in an upwards direction such that an upper end of the post is closer to the central vertical plane than a lower end of the post.
8. The mask of claim 7, wherein the angle of the post is such that a retaining force applied by the

headgear in use is substantially perpendicular to the angle of the post.

9. The mask frame of claim 1, wherein each of the first lateral arm and the second lateral arm extend: i. laterally outwardly from the central region of the mask frame; ii. rearwardly, towards an ear of the user; and iii. upwardly, so that each of the first lateral arm and the second lateral arm is angled upwards such that it extends along a direction configured in use to extend from the distal end of a respective one of the first lateral arm and the second lateral arm to an area between a temple and the ear of the user.

10. The mask frame of claim 9, wherein each of the first lateral arm and the second lateral arm extends upwardly along a vector configured in use to pass from below a nose of the user to a point between the temple and a top of the ear of the user.

11. The mask frame of claim 9, wherein each of the first lateral arm and the second lateral arm comprises a planar strip, an end of the planar strip defining a top corner and a bottom corner at the top margin and the bottom margin, respectively, wherein each of the first lateral arm and the second lateral arm twist along its length such that the bottom corner is positioned further away from the central region of the mask frame than the top corner.

12. The mask frame of claim 11, wherein each of the first lateral arm and the second lateral arm is tapered along its length such that a distance between the top margin and the bottom margin reduces along at least part of the length of the respective one of the first lateral arm and the second lateral arm.

13. The mask frame of claim 1, wherein the distal end of each of the first lateral arm and the second lateral arm is positioned below a notional horizontal mid plane that passes through the center of the conduit connection aperture.

14. The mask frame of claim 1, wherein the headgear connector comprises a rotation limiting formation configured to limit relative rotation between headgear connected to the headgear connector and the mask frame, wherein the rotation limiting formation comprises an end stop against which the headgear abuts after a predetermined amount of relative rotation between the mask frame and the headgear.

15. The mask frame of claim 1, wherein the acute angle is substantially 30°.

16. The mask frame of claim 1, wherein the aperture is rectangular in shape.
